

Security Groups and NACLs: Key Concepts

1. Security Groups:

- **Inbound Rules:** Define what traffic is allowed to reach your server. If no inbound rules are defined, no traffic is allowed into the server.
- **Outbound Rules:** By default, AWS allows all outbound traffic. You can configure these rules to restrict or allow traffic based on client requirements.
- **Ports:** It's essential to understand which ports need to be opened or restricted based on the services running on your instance. For example, port 80 is for HTTP, and port 443 is for HTTPS.
- **Tags:** Labels with key-value pairs that help in searching and organizing your resources.
- **Management:** You can delete inbound rules to stop services like HTTP from accessing the server. Security groups act as a shield, controlling who can access the server.

2. Network ACLs (NACLs):

- **Subnet-Level Security:** NACLs provide security at the subnet level, while security groups operate at the instance level.
- **Blocking Unauthorized Traffic:** If there is unauthorized access to a server via security groups, NACLs can be used to block traffic at the subnet level.
- **Stateful vs. Stateless:**
 - **Security Groups:** Stateful, meaning if incoming traffic is allowed, the corresponding outgoing traffic is automatically allowed.
 - **NACLs:** Stateless, meaning inbound and outbound rules need to be explicitly defined.
- **Numbering:** NACLs have rule numbering, where lower numbers have higher priority.
- **Denying Traffic:** NACLs offer the option to deny traffic explicitly, which is not available in security groups.

3. Elastic IP:

- **IP Address Management:** If a server is stopped, the public IP address will change when it is restarted. To avoid this, you can use an Elastic IP.
- **Static IP:** An Elastic IP is a static IP that does not change, even when the instance is stopped and started.
- **Purchase and Associate:** You need to allocate an Elastic IP from the AWS service and associate it with your instance using the "Associate Elastic IP address" option in the actions menu.